

RESEARCH ARTICLE

Key Considerations to be Applied While Leveraging Machine Learning for Financial Statement Fraud Detection: A Review

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ABSTRACT Financial statement fraud (FSF) is a challenging issue in capital markets and severely affects their overall health and stability. The effective prediction of FSF has become an urgent need for investors. In recent years, scholars have developed several machine learning-based FSF prediction models. This study conducted a systematic review of such models to facilitate an understanding of the latest developments in this field. First, sample and data preprocessing were analyzed, focusing on key aspects such as data sources, splitting training and testing sets, imbalanced samples, cross-period FSF, and handling of zero and missing values. Second, existing research on FSF prediction models was reviewed considering structured and unstructured data. The existing studies exhibited two significant characteristics: expansion if data from structured to unstructured formats and the evolution of methodologies from traditional machine learning to deep learning approaches. Third, the effectiveness of FSF prediction models was evaluated. Indiscriminately pursuing higher recall rates is not advisable. Rather, the effectiveness of the model in terms of predicting FSF must be scientifically assessed. Finally, the challenges and opportunities in current research were summarized. The core challenges were identified as the development of robust prediction models and the incorporation of unstructured data into these models. Moreover, leveraging diverse data, deep learning, and large language models can significantly enhance the performance of prediction models. Furthermore, to advance research in this field, this study advocates the construction of a shared and open FSF database.

INDEX TERMS Financial statement fraud, machine learning, deep learning, literature review.

I. INTRODUCTION

Financial fraud exhibits various forms based on the industry involved. References [1], [2], and [3] classified financial fraud into several categories: financial statement fraud (FSF), bank fraud, insurance fraud, and healthcare fraud. This study focused on FSF. According to the International Standard on Auditing 240 [4], FSF refers to the intentional misrepresentation or omission of material facts to mislead users regarding financial statements. Financial statement restatement is the reissuance of corrected financial statements by a company to address significant errors, omissions, or misleading information in previously issued reports [5]. Several studies have

frequently examined financial statement restatements when studying FSF. This is because of the similarities between financial statement restatement and FSF. Thus, to conduct a more in-depth investigation of FSF, this study included financial statement restatement within its purview.

FSF is a complex and complex topic in the capital market. It is known to affect investors and creditors, while also degrading the fairness and transparency of the capital market. Famous cases such as those of Enron, WorldCom, and Parmalat resulted in bankruptcy or the reorganization of these companies, thereby causing widespread harm to the entire market. For example, the Enron case resulted in the bankruptcy of a company with a market value of approximately \$60 billion and reduced the “big five” accounting firms to the “big four.” According to estimates by [6],

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approximately 10% of listed companies release FSF news every year, with losses accounting for approximately 1.6% of their annual equity value (equivalent to \$830 billion in 2021). These FSF situations are alarming because they significantly affect the stability and overall balance of the investment market and also erode owners' resources and undermine their trust. Moreover, accumulated FSF could result in economic recession. This because market participants rely on inaccurate financial information to make investment and production decisions [7]. The critical factor for preserving investor interests involves the successful prediction and identification of potential FSF in this environment. Thus, financial fraud forecast and identification research has useful significance.

Studies on FSF prediction exhibit two key features. First, structured to unstructured data have been added to the files that are analyzed via forecast models. Financial statements are a common example of organized data. They include different accounting items with clear layouts and specific spots for each. Unstructured data have been used to create FSF forecast designs in several studies in recent years. Management discussion and analysis (MD&A), earnings call transcripts, and audio/video recordings are included in unstructured data. Unstructured data analysis expands the scope and relevance of projection designs, thereby enabling the discovery of new, invisible fraudulent behaviors and identification of traditional styles of FSF. Second, the analytical methods have evolved from traditional machine learning to deep learning. Deep learning, which is a branch of machine learning, has garnered significant attention for its exceptional performance across areas such as image recognition and natural language processing (NLP). Deep learning is better suited for handling complex data patterns and relationships. It improves model prediction accuracy and generalization capabilities and has demonstrated greater potential and advantages in the field of FSF prediction.

Despite technological advancements providing new tools and methods for the prediction of FSF and generation of various research results, the practical applicability and effectiveness of these findings remain debatable. Regulatory agencies, accounting firms, analysts, and investors have shown great interest in FSF models. However, there is significant uncertainty regarding the development of FSF prediction in practice, the extent to which businesses rely on the models' predictions, and the effectiveness of these predictions. Reference [8] found that legal advisors at accounting firms advise against the implementation of FSF prediction models in audit practices. They argue that plaintiff lawyers may exploit this implementation to accuse auditors of negligence for not appropriately addressing fraud risks in financial statements. Therefore, this study aimed to summarize and analyze the current research findings on FSF prediction. The entire process of FSF prediction models, including sample selection and data preprocessing, model construction, and performance evaluation, was elucidated.

The contributions of this study are as follows. This review focused on the projection of FSF and provided a more

thorough research. Reference [9] reviewed financial fraud prediction studies across 2004–2015, and [10] covered studies within 2009–2019. These studies included FSF, bank fraud, insurance fraud, healthcare fraud, etc. In contrast to other studies, this study focused on predicting FSF. It examined important aspects such as handling samples and data, building prediction models, and evaluating their performance. Thus, this study provides specialized insights for this area.

Second, [11] thoroughly reviewed FSF. They highlighted that future research should focus on analyzing unstructured data, such as MD&A, earnings call transcripts, and media sources such as audio and video. This study built on their previous work by including these unstructured data resources into FSF forecast models, thereby providing a more in-depth scientific foundation.

Third, this study integrated expertise from the fields of finance and accounting considering algorithms from computer science to develop a comprehensive framework. This framework is a significant theoretical contribution to FSF prediction research and provides practical, actionable technical support for professionals engaged in FSF detection.

The remainder of this paper is organized as follows. Section II presents specimen collection and information processing in FSF forecast models. Section III describes and evaluates reports that create FSF forecast models based on structured data. Unstructured data used to create FSF forecast models are presented in Section IV. Section V presents an assessment of efficacy of these concepts. Section VI discusses the challenges and prospects in current research. The specific activities that industry associations should follow are described in Section VII. Finally, Section VIII concludes this paper.

II. SAMPLE AND DATA PREPROCESSING

Samples are the foundation for constructing prediction models. Consequently, the accuracy and completeness of records and data play a crucial role in model performance. Data preprocessing is crucial in building models. It must be precise, detailed, and strict. This study focused on five topics: data origins, splitting training and test sets, unequal fraud samples, fraud over multiple periods, and handling zeros and missing data.

A. DATA SOURCES

Marking fraud samples in financial statements facilitates the creation of prediction models. Splitting samples into fraud and normal boosts model performance and adaptability. References [12] and [13] listed the popular databases of fraud and restated financial statements of public firms in the United States of America. The four databases considered were as follows. (1) Accounting and auditing enforcement releases (AAER), compiled by the center for financial reporting and management at the University of California, Berkeley. (2) Government accountability office. (3) Audit analytics (AA). (4) The Stanford securities class action clearinghouse.

These databases vary in terms of their method of collecting and organizing data. Each offers unique benefits and drawbacks. AAER stands out as all its samples are reviewed by the Securities and Exchange Commission (SEC). This renders the data objective, thereby, it is easier for scholars to replicate empirical results. However, AAER only covers FSF investigated and sanctioned by the SEC. It ignores frauds and restatements not examined by the SEC. The GAO includes restatement data but only for the first year. It does not mention the full restatement period. Reference [12] reported that different directories have different performance levels for prediction models. Thus, creating a unified fraud database with consistent standards is key. It ensures that the models are strong and reliable. Moreover, mistakes from various data can be eliminated by utilizing unified requirements and data sources, thereby improving the reliability and security of prediction models. Furthermore, it is easier to compare results horizontally.

Several studies have utilized FSF data from companies in various countries. References [14], [15], and [16] used fraud datasets from China. Reference [17] analyzed data from Iran. Reference [18] studied the data from Vietnam and [19] focused on the United Arab Emirates (UAE). Furthermore, [20] considered fraud data from Greece.

B. DIVIDING TRAINING AND TEST SETS

When developing FSF prediction models, it is customary to utilize both a training dataset (incorporating a validation subset) and a separate testing dataset. The training set is employed to construct the projection unit and determine its parameters, while the test set is used to assess the effectiveness and generalization capabilities of the forecasting model. Reference [21] sourced their training dataset for the period 1991–2000 and utilized samples from 2001–2003 for testing purposes. Considering that FSF often necessitates approximately two years to be detected [22], [23] introduced a 24-month interval between their training and testing datasets. They trained their model on data spanning 1991–2001 and evaluated it using samples from 2003.

C. IMBALANCED FSF SAMPLES

The insufficient number of fraud samples poses significant challenges for the development of models. Reference [23] analyzed 206,026 cases and classified only 1,171 (0.57% of the total) as FSF samples. If FSF samples forms a tiny fraction, classification approaches must evaluate numerous probable patterns. Insufficient fraud samples obscure the distinction between genuine patterns and noise, thereby increasing the likelihood of overfitting in classification models. Even advanced models aimed may still mistakenly label several fraudulent instances under such circumstances. The large imbalance in the count of fraudulent enterprises versus explanatory factors can impair the model's ability to predict future cases, thereby leading to weak generalization [24].

Previous studies have employed four methods to solve this problem. The first involves improving the sampling methods for FSF samples [24]. The second is the consideration of a different, more prevalent event that is directly related to FSF [25]. The third is the adjustment of the misclassification costs [26]. Finally the fourth is the application of generative adversarial networks (GAN) to create artificial fraud samples [27]. The costs of misclassification are elucidated in Section VI.

Reference [24] thoroughly examined various sampling techniques and evaluated their respective performance. Multi-subset observation undersampling (OU) addresses the imbalance between fraud and regular tests by generating multiple replicas of the original data. Here, each replica encompasses all instances of fraud coupled with differing randomly selected sets of regular tests. Multivariable undersampling (VU) generates a different set of randomly selected input variables, thereby correcting the mismatch between fraud samples and descriptive attributes. The third method involves the classification of each fraud type using multivariable partitioned variable undersampling (PVU). It uses existing knowledge to classify variables into different subsets based on their association with specific fraud categories. Reference [24] reported that the OU and PVU methods reduce misclassification cost by 10.8% and 9.6%, respectively, whereas the VU method improves fraud detection in certain situations. Reference [28] proposed three feature selection methods, including reducing overlap without sampling.

When a company makes significant accounting errors, it usually conducts a financial restatement. Compared to FSF, there are more abundant samples of financial restatements. This facilitates earlier and more accurate predictions of financial errors. Reference [25] studied a sample of 54,354 company-years, with 3,599 company-years involving financial restatements (proportion of 6.62%), which is considerably higher than that of FSF. With the evolution of FSF and the increasing complexity of hidden fraudulent activities, excluding unintentional financial restatement samples may results in incomplete prediction models. Focusing solely on intentional financial restatements might in a misleading manner suggests that unintentional financial restatements (errors) are unimportant for companies and their investors. In fact, the market reacts similarly to both unintentional and intentional financial restatements, consequently eroding investor confidence. Moreover, unintentional misstatements are more likely caused by weak internal controls, occur more frequently, and exert a broader impact [17]. References [13] and [29] differentiated between intentional and unintentional financial restatements and developed financial restatement prediction models.

GAN (a deep learning model) is structured with a generator and a discriminator. It can learn the data distribution and generate high-quality new samples that are indistinguishable from real samples. By using GAN to generate simulated fraud samples, the number of fraud samples can be effectively

increased, thereby improving data balance and mitigating the problem of data imbalance to a certain extent. Reference [27] applied GAN to address the challenges posed by unequal distribution of FSF samples. Reference [30] developed a synthetic data generation methodology based on GAN. They constructed data following an analysis of the characteristics of the original dataset.

D. CROSS-PERIOD OF FSF

FSF has the potential to cover several successive periods, leading to a chain of fraudulent behavior. The average duration of FSF is 3.1 years [31]. The model may employ the same features to predict, when the companies commit fraud both during the training and testing periods [23]. However, this processing may result in data leakage. Data leakage occurs when the model training is delayed until the testing period, thereby yielding an overly optimistic testing result. This method primarily employs testing data to train the model, rather than using training features. The performance of the model in real-world scenarios remains unsatisfactory.

References [23] and [24] attempted to address this problem. Reference [24] omitted samples related to multiple fraud reporting for the same companies. Consequently, they used only the first-year FSF samples. Whereas, [23] reset all fraud samples to zero during the training period. These techniques can lessen the risk of data leakage.

E. HANDLING ZERO VALUES AND MISSING VALUES

Zero values and missing values are frequently required during the creation of eff prediction models. Zero values represent cases wherein the data is zero. In contrast, missing values indicate the absence of data or information.

Reference [21] noted that zero values in accounting items can result in the denominator becoming zero during kernel function calculations. To prevent this, studies often replace these zeros with a tiny number (e.g., 0.0001), instead of zero itself.

Various approaches have been employed in the writing to handle missing values. If the percentage of missing values of a model exceeds a certain threshold, one approach involves the removal of variables. References [21] and [23] removed the variables with more than 25% missing values. Another strategy involves imputing missing values appropriately. Reference [18] filled missing values with randomly generated numbers from a standard normal distribution. They reported that random numbers prevented inconsistencies with the assumed distribution, in contrast to fixed values such as 0. Different methods for addressing missing values have their advantages and disadvantages. The best choice depends on the data features and model requirements.

III. CONSTRUCTION OF FSF PREDICTION MODELS USING STRUCTURED DATA

In studies constructing FSF prediction models based on structured data, scholars have primarily analyzed the application of different machine learning methods. These methods

include logistic regression, support vector machines (SVM), ensemble learning, neural networks (NN), and deep learning. Notably these machine learning methods are also applicable for yeh construction of prediction models based on unstructured data. This study did not exhaustively cover all machine learning methods but focused on the commonly used algorithms.

When constructing FSF prediction models, scholars typically use multiple machine learning methods simultaneously. Consequently, they compare their performance in building prediction models. Owing to differences in data sources, data preprocessing methods, and performance evaluation metrics used in the literature, the evaluation results of the models (i.e., the advantages and disadvantages of machine learning methods) may vary and even be contradictory. However, the overall trend indicates the gradual replacement of traditional machine learning methods with more advanced algorithms such as ensemble learning and deep learning. These emerging algorithms excel at managing intricate data patterns and correlations, thereby enhancing the predictive accuracy and generalization capacity of the models. Thus, they have showcased their remarkable potential and benefits in the realm of FSF forecasting. Table 1 presents an overview of the literature discussed in this section.

A. LOGISTIC REGRESSION

A machine learning algorithm that relies on the logistic function (the Sigmoid function) is logistic regression. It establishes the relationship between input features and output probabilities to achieve classification predictions. Reference [9] summarized research on the detection of financial fraud across 2004–2015. They reported that approximately 13% of the literature used logistic regression algorithms. Logistic regression is intuitive and can easily identify the characteristics of fraudulent companies and their marginal effects. Thus, it aids scholars in better understanding the influencing factors and mechanisms of FSF. For example, [31] reported that the probability of FSF increased by 0.52% for loss-making companies in that year.

B. SVM

SVM involves determining a hyperplane in the sample space that effectively separates samples of different classes, considering a training set $S = \{(u^1, y^1), \dots, (u^l, y^l)\}$ ($y_i \in \{-1, +1\}$, with y_i indicating negative and positive examples). This hyperplane can be described as $\{x: < w, x > + b = 0\}$, which can accurately separate positive and negative examples. In real situations, the feature space is often nonlinearly separable. In such cases, SVM can employ kernel functions to map the low-dimensional space to a high-dimensional space, thereby converting a nonlinearly separable space into a linearly separable space. Reference [10] analyzed the literature on the detection of financial fraud across 2009–2019 and found that SVM was among the most commonly used methods (approximately 23% of all studies). Reference [32] compared logistic regression, SVM, decision trees, ensemble

TABLE 1. Summary of the construction of FSF prediction models using structured data.

Ref.	Publication Year	Dataset	Algorithms	Key Findings
[13]	2017	United States (AA)	SVM, NN	The study distinguishes between intentional (fraudulent) and unintentional (error) financial restatements. The findings reveal that NN outperform other algorithms.
[14]	2022	China	Meta-Learning	The study found that the use of a meta-classifier improved model performance.
[15]	2022	China	Logistic, RF, XGBoost	The study constructed a related transaction knowledge graph from the network of related transactions.
[16]	2023	China	Logistic, SVM, RF, GBDT, RUSBoost, NN	Fraud-related variables were identified and organized within the GONE framework.
[19]	2021	United Arab Emirates	Logistic, SVM, NN	The SVM achieved 89.54% classification accuracy, outperforming classifiers like Logistic and NN.
[20]	2019	Greece	NN	The newly proposed predictive model was tested using data from Greek companies, demonstrating strong robustness and promising applicability.
[21]	2010	United States (AAER)	SVM	The SVM is proposed to detect FSF using basic financial data.
[23]	2020	United States (AAER)	SVM, RUSBoost	Utilizing both raw financial data and the RUSBoost algorithm can enhance the model's performance.
[25]	2021	United States (AA)	SVM, GBRT	Accounting variables alone are insufficient for detecting misstatements, but combining them with audit and market variables improves model performance.
[26]	2022	United States (AAER)	RUSBoost, Cost-sensitive Cascading Forest	The model proposed in the paper demonstrates performance that exceeds that of the baseline model.
[31]	2020	United States	Logistic, SVM, RF, NN	The logistic and SVM perform well. Estimate the expected ex-ante costs of financial misrepresentation.
[32]	2011	United States (AAER)	Logistic, SVM, RF, NN	Logistic and SVM have demonstrated strong performance.
[33]	2017	United States	SVM, NN	By implementing feature selection, the SVM algorithm's accuracy was boosted to 88.37%.
[36]	2023	China	RF, AdaBoost, GBDT, XGBoost, LightGBM	The results showed that bagging outperformed boosting in all evaluation metrics.
[37]	2018	China	Logistic, SVM, RF, NN	The performance of the RF surpassed that of other methods.
[39]	2011	China	Logistic, SVM, NN	The study results showed that without feature selection, NN had the best prediction performance; after feature selection, NN still performed well.
[40]	2024	China	NN	The study proposed a Transformer-based distributed distillation architecture for financial fraud detection.
[41]	2012	United States (AAER)	Meta-Learning	The study found dynamic meta-models greatly outperformed baseline models in prediction.
[42]	2022	United States (AAER)	BAS	The study proposes an optimization method using a meta-algorithm integrated with BAS.
[43]	2015	United States (AAER)	FSD score model	The study found that higher FSD scores reflect a higher degree of financial statement errors.

TABLE 1. (Continued.) Summary of the construction of FSF prediction models using structured data.

Ref.	Publication Year	Dataset	Algorithms	Key Findings
[44]	2019	United States (AAER)	ABF score model	The study combined Benford's Law and F-score model [45] to create the ABF score model.
[46]	2022	China	SVM, RF	The study designed a managerial knowledge graph based on links between companies, executives, and related institutions.
[47]	2022	China	PE-EDD	The study incorporates behavioral patterns and peer effects among firms into its predictive analysis.

Note: The dataset column includes the names of the databases in parentheses. The algorithms column lists only the algorithms mentioned in this paper; in some studies, not all algorithms are included.

learning, and NN, and found that logistic regression and SVM algorithms performed well in financial fraud prediction.

Reference [21] used SVM to construct FSF prediction models. They designed the financial kernel to address the problem of nonlinear separability in the feature space. The financial kernel generated $3n(n-1)$ features based on n attributes. These features can be divided into six types, representing the ratios within a year and the year-over-year changes in these ratios. The fundamental concept of the financial kernel is that to hide FSF, companies manipulate different accounting items. These items can be single numbers in financial statements, combinations of numbers, or even numbers that change over time. Reference [21] first collected 40 accounting items, excluded those with more than 25% missing values, and ultimately retained 23 accounting items, thereby generating 1,518 features. The study results indicated that the optimal model achieved an 80% recall rate for FSF and a 90.6% for nonfraud cases. Reference [33] found that feature selection improved the classification accuracy of SVM to 88.37%. Reference [19] used different methods to predict FSF in UAE companies, with the SVM classification accuracy of 89.54%.

C. ENSEMBLE LEARNING

Ensemble learning is a machine learning approach that integrates multiple learners in a particular manner. By amalgamating these learners, Ensemble learning exhibits notably superior performance compared to a solitary learner. However, this performance improvement often necessitates higher computational demands and yields less interpretable prediction results [34].

Bagging and boosting are two important branches of ensemble learning algorithms [35]. Bagging is a parallel ensemble method that generates multiple subsamples by sampling the training set with replacement. Each subsample trains a base model, and the final prediction is obtained by averaging or voting on the predictions obtained from all base models. Boosting is a sequential ensemble method that iteratively trains multiple weak learners. Herein, each learner adjusts the sample weights based on the performance of the

previous model. Thus, the new model is more focused on the samples that were previously mispredicted. Both algorithms offer their advantages and disadvantages, and the choice between them should be based on specific circumstances. Reference [36] compared the performance of ensemble learning algorithms based on bagging and boosting. Bagging outperformed boosting in all evaluation metrics. However, owing to differences in the samples, feature variables, and algorithms used in the study, these results necessitate further discussion and validation.

1) ENSEMBLE LEARNING ALGORITHMS BASED ON BAGGING

Random forest is an ensemble learning algorithm based on the concept of bagging. When generating each decision tree, random forest randomly samples both the data and features, thereby increasing the diversity of the base learners. This randomness results in random forest exhibiting performance and robustness. Thus, it typically excels in handling high-dimensional data and large datasets.

Reference [16] identified a set of variables related to FSF and organized them into different categories within the GONE framework (greed, opportunity, need, and exposure). They used various machine learning algorithms for prediction and observed that random forest outperformed other algorithms in terms of predicting FSF. Reference [37] showed that random forest performed better than other algorithms. Reference [26] proposed the concept of cost-sensitive cascading forest, which is an algorithm that adaptively adjusts the forest depth based on recall rate. It assigns greater weight to minority classes to penalize misclassification of these classes.

2) ENSEMBLE LEARNING ALGORITHMS BASED ON BOOSTING

Adaptive boosting (AdaBoost) is a classic boosting algorithm. Its core idea involves iteratively training multiple weak learners, adjusting the sample weights based on the performance of the previous model in each iteration, and finally focusing on misclassified samples to improve the overall classifier's performance. Random undersampling

boosting (RUSBoost) is an improvement on AdaBoost. It introduces random undersampling. During each iteration, RUSBoost balances positive and negative samples by adjusting sample weights and then randomly selecting a subset of majority class samples. Reference [23] employed the RUSBoost algorithm to build FSF prediction models. However, incomplete theories may result in financial ratios losing information. Thus, the use of using raw data for prediction models could be more beneficial. They fed raw financial data into a prediction model using RUSBoost and observed that combining it with ensemble learning enhanced the model performance.

Representative boosting algorithms include gradient boosting decision trees (GBDT), light gradient boosting machine (LightGBM), and eXtreme gradient boosting (XGBoost). GBDT reduces the loss function by training multiple decision trees via gradient descent. Each new tree fits the residuals of the previous one, and the final prediction sums all the predictions of the model. LightGBM enhances GBDT for high-dimensional features and large datasets. Further, XGBoost trains multiple weak learners iteratively on residuals, thereby combining predictions for a stronger model. In addition, it uses L1 and L2 regularization to manage complexity and avoid overfitting.

Reference [25] created a dataset with over 100 features. They incorporated financials, audits, credit ratings, and corporate governance. They applied GBRT to develop a FSF prediction model. The top 10 (20) feature variables accounted for 26.27% (47.42%) of the total impact. The top three feature variables that influenced the prediction model were soft assets, bid-ask spread, and nonaudit fee ratio. Notably, accrual variables commonly used in accounting research were not prominent in the model, ranking only 20th.

D. NN AND DEEP LEARNING

The development of NN can be traced back to the 1950s; however, they have gained widespread attention and application only in the past decade owing to the rise of deep learning. Inspired by the human nervous system, NN are machine learning algorithms comprising multiple neurons (or nodes) organized into several layers, including input, hidden, and output layers. These neurons transmit information through connection weights and achieve nonlinear mapping through activation functions. Thus they enable NN to learn and model complex, high-dimensional data relationships. Reference [38] reported that NN can handle complex, high-dimensional, and nonlinear relationships, thereby effectively overcoming the curse of dimensionality.

Reference [39] used multilayer feedforward NN, probabilistic NN, SVM, and logistic regression algorithms to construct FSF prediction models. They showed that without feature selection, probabilistic NN exhibited the best prediction performance. Moreover, after feature selection, the probabilistic NN still performed well. Reference [13] distinguished between intentional (fraud) and unintentional

(error) financial restatements and constructed corresponding prediction models. They found that the NN yielded the best results. Reference [20] introduced a fraud detection method using a deep dense neural network. They tested it with Greek companies' data, and found it to be robust and promising. Reference [40] suggested a Transformer-based distributed knowledge sharing system for detecting FSF. They aimed to solve data inconsistencies and imbalance across industries.

Various NN such as convolutional NN, recursive networks, and autoencoders have been created by researchers. Deep learning enables the capture of intricate patterns and features, facilitating advanced tasks. Deep learning has made remarkable advances in NLP, image recognition, and speech recognition, and has become a key artificial intelligence technology today. Section VI further explores the application of deep learning in this field.

E. OTHER MACHINE LEARNING ALGORITHMS

Meta-learning involves learning by automating tasks such as selecting algorithms and optimizing hyperparameters. A meta-learner gains general knowledge from past tasks, and applies it to new tasks to accelerate learning or boost performance. Reference [41] applied meta-learning to enhance FSF predictions by learning from various tasks. They found that dynamic meta-models outperformed baselines. Reference [14] used meta-classifiers to boost model performance.

Reference [42] proposed the application of the Beetle Antennae Search (BAS) algorithm to FSF prediction models. BAS, inspired by a beetle's antennae searching for food, simulates their natural searching process.

F. OTHER STRUCTURED DATA METRICS

Researchers have constructed FSF prediction models based on the distribution of financial statement figures. Reference [43] created the FSD score to measure deviations from Benford's Law in financial statements. Higher FSD scores indicate more financial errors. Reference [44] merged Benford's Law with F-score model [45] to form the ABF score model, which was found to outperform others.

Scholars construct FSF prediction models using financial numbers, ratios, and other structured data from annual reports. Reference [46] created a managerial knowledge graph. They mapped relationships among listed companies, executives, and institutions. Reference [15] built a knowledge graph of related transactions based on transaction networks. Reference [47] introduced a prediction model based on error distribution (PE-EDD). This model combined firm behavior and peer effects to conduct predictive analysis based on error distribution and peer effects. By enhancing feature dimensions, these studies have rendered FSF prediction models more capable of detecting signs and risks comprehensively.

IV. CONSTRUCTION OF FSF PREDICTION MODELS USING UNSTRUCTURED DATA

Recently, deep learning advancements have resulted in researchers exploring unstructured data for FSF prediction

model building. Unstructured data, such as MD&A, earnings call transcripts, and audio/video recordings, is being used for FSF prediction. The use of unstructured data boosts fraud prediction accuracy and efficiency, despite increasing the model complexity and computational costs, by offering more comprehensive information. The analysis of this data reveals both traditional FSF patterns and hidden fraudulent behaviors, thereby enhancing the prediction models' applicability and practicality. Reference [11] reported that future research would leverage text and audio data for building predictive models. Table 2 presents an overview of the literature discussed in this section.

A. CONSTRUCTION OF FSF PREDICTION MODELS USING TEXT DATA

When a company engages in FSF, the writing style used to convey financial information often changes significantly [48]. Reference [49] summarized the theoretical frameworks and methodologies for analyzing FSF texts. They considered content-based criteria analysis, scientific content analysis, reality monitoring, the hypothesis of managerial obfuscation, the theory of information manipulation, the theory of interpersonal deception, the four-factor theory, and the leakage theory.

Text data includes MD&A and earnings call transcripts. In terms of studies that have constructed FSF prediction models based on MD&A, [50] showed that the text in annual reports could be used to predict FSF. References [51] and [52] constructed dictionaries based on MD&A and used SVM to construct prediction models. Reference [53] integrated financial statement disclosure topics into prediction models based on financial data and text. Consequently, they significantly improved the prediction of financial misreporting by 59%.

In terms of studies that have predicted FSF prediction models based on earnings call transcripts, [54] suggested that executives were aware of whether the financial statements were manipulated. The spontaneous words of executives provide clues that can be used to identify fraudulent behavior. They labeled the narratives of earnings call transcripts as fraudulent using different methods, extracted linguistic features using machine learning methods, and trained prediction models. It was observed that the performance of the language classification model based on CEOs' speech significantly outperformed random guessing, improving by 6%–16%. Whereas, deceptive CEOs and CFOs tended to use general knowledge, fewer non-extreme positive emotion words, and mentioned shareholder value less frequently in their narratives.

Deep learning is crucial for text analysis and prediction model construction. Convolutional networks exhibit excellent performance in semantic segmentation [55]. Reference [56] found that HAN exhibited excellent predictive performance and could capture the structured hierarchy of documents, thereby providing word and sentence-level early warning recognition. Reference [57] discovered that long short-term

memory (LSTM) and gated recurrent unit (GRU) deep learning algorithms significantly improved model performance and yielded good classification results. Reference [58] proposed an attention-based multimodal deep learning algorithm (RCMA) that integrated fine-grained attention mechanisms, including ratio-aware attention, section-aware attention, and modality-aware attention mechanisms. The proposed algorithm outperformed benchmark models in terms of predicting FSF.

B. CONSTRUCTION OF FSF PREDICTION MODELS USING AUDIO DATA

Research in social psychology has shown that fraudsters' emotional and cognitive processes may produce many markers that can aid in the identification of fraud. These markers include verbal cues (e.g., speech content), non-verbal cues (e.g., tone, facial expressions, or gestures), and physiological changes (e.g., heart rate). Scholars often employ layered voice analysis (LVA) technology to analyze audio data. LVA comprises a set of signal processing algorithms that extract and combine attributes from speech to identify different types of stress, cognitive processes, and emotional responses. Its primary goal involves measuring various emotions and determining whether speech segments are deceptive.

LVA comprises three output layers. The basic layer extracts and combines raw sound attributes. The second layer forms fundamental emotional indicators. Finally, the third layer draws conclusions based on the combination of emotions from the previous layer. Different commercial contexts have different LVA applications. Ex-Sense Pro R software is an LVA analysis application that measures speech waveforms. It analyzes deviations of the speaker from the baseline to produce four vocal characteristics: emotional stress, cognition, general stress, and thinking levels.

Reference [59] used Ex-Sense Pro R software to extract vocal features and found that the disharmonious voice of managers during earnings calls was positively correlated with the likelihood of restatements owing to violations. Reference [60] combined financial figures, textual data, and vocal features to train a FSF prediction model. The combination of features across categories provided better predictive performance than that when using a single category of features alone. Reference [61] noted that companies have begun to regulate word usage based on perceived sentiment, thereby rendering traditional text analysis unreliable. They proposed a multimodal bidirectional LSTM framework, which enhanced the analysis of earnings call audio and text data through a cross-attention fusion mechanism. Reference [62] proposed a deep learning-based video analysis system for automated compliance auditing in stock brokerage.

V. EVALUATION OF THE EFFECTIVENESS OF FSF PREDICTION MODELS

The evaluation of the effectiveness of FSF prediction models is extremely important. The correct assessment of a model's performance facilitates its improvement and ensures that

TABLE 2. Summary of the construction of FSF prediction models using unstructured data.

Ref.	Publication Year	Dataset	Data Type	Algorithms	Key Findings
[50]	2017	United States (AAER)	MD&A	Logistic, SVM, NN,	The study results showed that the text in annual reports can be used to predict financial fraud.
[51]	2010	United States	MD&A	SVM	The study showed that MD&A text enhances the predictive power of quantitative financial data for bankruptcy and fraud detection.
[52]	2015	United States (AAER)	MD&A	SVM, RF	The study constructed dictionaries based on MD&A and used SVM to build prediction models.
[53]	2020	United States (AAER, AA)	MD&A	LDA	The study integrated financial statement disclosure topics into prediction models based on financial data and text.
[54]	2012	United States	Earnings call transcripts	Logistic	The language classification model based on CEOs' speech significantly outperformed baseline methods.
[56]	2020	United States	MD&A	Logistic, SVM, XGBoost, NN, HAN	The study found that HAN not only have excellent predictive performance, but also capture the structured hierarchy of documents.
[57]	2022	China	MD&A	LSTM, GRU	The study found that the LSTM and GRU methods achieved correct classification rates of 94.98% and 94.62%.
[58]	2023	China	MD&A	RCMA	RCMA integrates fine-grained attention mechanisms and demonstrates strong predictive performance.
[59]	2012	United States	Earnings call voice	Logistic	The study found a positive correlation between managers' disharmonious tone during earnings calls and restatements.
[60]	2015	United States (AA)	Earnings call transcripts and voice	Logistic, Naive Bayes, Generalized likelihood ratio test based on Bayesian	The results showed that combining features across categories improved predictive performance over using a single category.
[61]	2022	United States	Earnings call transcripts and voice	LSTM	The LSTM model enhances the analysis of earnings call audio and text data through a cross-attention fusion mechanism.

Note: The dataset column includes the names of the databases in parentheses. The algorithms column lists only the algorithms mentioned in this paper; in some studies, not all algorithms are included.

the model can reliably perform its intended function when deployed. The evaluation of the effectiveness of a model is both the endpoint of constructing a FSF prediction model and the logical starting point for its application. However, the accuracy of predicting FSF is not always better with higher values. This is because increasing the success rate of models in the detection of FSF increase the number of false positives. Even the best models exhibit a false positive rate to recall rate ratio exceeding 100:1 [8]. An increase in the number of

false positives incurs higher costs for manual investigations, thereby significantly increasing the cost of using the model. Therefore, the indiscriminate pursuit of higher recall rates is not advisable. The effectiveness of FSF prediction models must be scientifically evaluated.

A. COMMON EVALUATION METRICS

The task of FSF prediction constitutes a typical instance of binary classification. Depending on the actual and predicted

classes of the samples, it can be categorized into four distinct groups: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN), which collectively organize into a confusion matrix. Fraud and normal refer to the prediction results of the model, whereas true and false indicate whether the prediction matches the actual situation. Therefore, FP refer to cases wherein the model predicts a company as fraudulent, although it is actually normal. Whereas, FN refer to cases wherein the model predicts a company as normal, despite it being actually fraudulent. The confusion matrix is presented in Table 3.

TABLE 3. Confusion matrix.

Actual Situation	Predicted Results	
	Positive (P)	Negative (N)
Positive (T)	True Positive (TP)	False Negative (FN)
Negative (F)	False Positive (FP)	True Negative (TN)

Utilizing the confusion matrix, various performance metrics can be derived to evaluate the model. These include accuracy, precision, recall, false positive rate, and false negative rate. These metrics offer insights into the performance of the model considering diverse perspectives. Consequently, they aid in the realization of a comprehensive understanding of its efficacy under various scenarios.

The accuracy metric is determined by dividing the count of accurate predictions by the total number of samples evaluated. However, in imbalanced samples, only accuracy is not sufficient for a comprehensive measurement of the effectiveness of the model. For example, consider that in a certain dataset, only 5% of the companies are actually fraudulent. A prediction model developed based on this dataset may exhibit an accuracy of 95% following testing. On the surface, this accuracy appears high. However, if we do not use the model and simply predict all samples as normal companies, we would still achieve a 95% accuracy. Thus, this prediction model does not offer any performance improvement.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN}) \quad (1)$$

Precision and recall are important metrics for evaluating the performance of FSF prediction models. Precision measures the accuracy of the model in terms of predicting positive cases, that is, the proportion of actual fraudulent companies among those predicted as fraudulent. Further, recall measures the ability of a model to identify positive cases, that is, the proportion of correctly predicted fraudulent companies among all actual fraudulent companies. The false positive and false negative rates evaluate the error rate of the model on negative samples. From a statistical perspective, the false positive rate corresponds to the probability of a Type I error. Whereas, the false negative rate corresponds to the probability of a Type II error. These metrics are interrelated. Precision and recall as well as the false positive and false negative rates

typically exhibit inverse relationships. Reference [8] reported that most existing research on FSF prediction models has focused on recall while neglecting the importance of the false positive rate. The correct understanding and evaluation of these metrics is crucial for comprehensively understanding the performance of FSF prediction models.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad (2)$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad (3)$$

$$\text{False Positive Rate} = \text{FP} / (\text{FP} + \text{TN}) \quad (4)$$

$$\text{False Negative Rate} = \text{FN} / (\text{TP} + \text{TN}) \quad (5)$$

Under the same machine learning model, the prediction results for samples are known to vary. This is because machine learning typically generates a prediction value (or probability) for FSF for each sample, which is consequently compared to a specific threshold. If the prediction value exceeds the threshold, the sample is classified as fraudulent (P); otherwise, it is classified as normal (N). Therefore, scholars must set the threshold. Moreover, different thresholds may yield different prediction results for the samples, thereby affecting the related model evaluation metrics.

Consider the study by [45] as an example. They constructed an FSF prediction model and set an F-score threshold. If a sample's FSF prediction value exceeded the F-score threshold, it was classified as a fraudulent company; otherwise, it was classified as a normal company. Table 4 presents the relationship between the F-score threshold, false positive rate, and false negative rate. First, let us analyze the changes in the false positive rate with the F-score threshold. When the F-score threshold was 1, 36.3% of the normal companies were incorrectly classified as fraudulent companies. Whereas, for an F-score threshold of 2, this proportion decreased to 10%. Further, when the F-score threshold increased to 2.45, only 5% of the normal companies were misclassified as fraudulent companies. Next, we analyze the changes in the false negative rate with the F-score threshold. When the F-score threshold was 1, 31.4% of the fraudulent companies were incorrectly classified as normal companies. Further, for an F-score threshold of 2, this proportion increased to 68.2%. Finally, when the F-score threshold reached 2.45, 81.2% of the fraudulent companies were classified as normal companies. Thus, as the F-score threshold increased, the threshold for classifying samples as fraudulent companies also increased. Therefore, the probability of normal companies being incorrectly classified as fraudulent companies decreased, resulting in a lower false positive rate. Simultaneously, the probability of fraudulent companies being misclassified as normal companies increased, resulting in a higher false negative rate. Overall, the false positive and false negative rates exhibited opposite trends.

The threshold setting should not simply be set as high or low; rather, it should be balanced according to research and practical requirements. Assume that regulatory agencies use FSF prediction models and set a high F-score threshold. In that case, they will almost always accurately identify the

TABLE 4. Relationship between f-score threshold, false positive rate, and false negative rate.

F-score Threshold	False Positive Rate	False Negative Rate
1	36.3%	31.4%
2	10%	68.2%
2.45	5%	81.2%

Data from Fig. 2 in [45].

fraudulent companies at the cost of missing several fraudulent companies. Conversely, setting a lower F-score threshold will detect the maximum number of fraudulent companies at the cost of having a high number of samples marked as fraudulent, thereby leading to high investigation costs.

To facilitate a comprehensive understanding of the prediction performance of a model at different thresholds, [63] proposed the receiver operating characteristic (ROC) curve, which has been widely applied. The horizontal axis of this curve represents the true positive rate (i.e., recall), and the vertical axis represents the false positive rate. By determining the classification results of samples at different thresholds, calculating the true positive and false positive rates, and plotting these as a series of coordinates, the ROC curve is formed by connecting these points. The ROC curve typically lies above the line connecting points (0, 0) and (1, 1). This indicates that the performance of the model is better than random guessing. In general, if the ROC curve of one prediction model completely envelops that of another, the former is considered to perform better. The area beneath the ROC curve, which is referred to as area under the curve (AUC), indicates the region enclosed by the ROC curve and the x-axis. A higher AUC value implies superior overall model performance. The AUC of the area enclosed by the line connecting points (0, 0) and (1, 1) and the horizontal axis is 0.5; therefore, the AUC of a prediction model must not be lower than 0.5.

Reference [64] proposed the use of $NDCG@k$ for evaluating the performance of prediction models. $NDCG$ stands for normalized discounted cumulative gain. It is often used in assessing algorithms such as web search engines. First, the discounted cumulative gain at the top k positions ($DCG@k$) is calculated. In the ranking list, if the sample is fraudulent, rel_i equals 1, otherwise it is 0. Here, k represents the top k with the highest predicted probabilities in the test set. Second, the $NDCG@k$ is calculated. $NDCG@k$ refers to the value when all actual fraudulent samples are at the top of the ranking list. In addition, certain scholars have used other evaluation metrics, such as $Precision@k$ and $Recall@k$ [16].

B. MISCLASSIFICATION COST

Although metrics such as AUC, $NDCG@k$, $Precision@k$, and $Recall@k$ are commonly used for the comprehensive evaluation of the performance of machine learning models, these metrics treat FP and FN as incurring the same cost. Different prediction errors can result in different consequences.

For example, misclassifying a fraudulent sample as normal and a normal sample as fraudulent can exert different impacts. To balance the losses from the different types of errors, misclassifications can be assigned unequal costs. A cost curve can be used to reflect the overall error cost of the model. Consequently, different types of error costs can be adjusted to address the issue of imbalanced FSF samples. Reference [21] analyzed the performance of models under different misclassification costs. They found that the model achieved optimal performance when the cost ratio of misclassifying a sample as fraudulent to normal was set as 50:1. Reference [65] proposed cost-sensitive learning to address class imbalance and asymmetric misclassification costs.

The consequences of different types of misclassifications vary. Different decision-makers face different consequences depending on these errors. For example, for regulatory agencies, the misclassification of a normal company as a fraudulent one could result in unnecessary investigative measures, thereby wasting investigation costs. Whereas, the misclassification of a fraudulent company as a normal one could, in case of FSF issue, erode the market confidence, thereby resulting in more severe consequences. For auditors, the misclassification of a normal company as fraudulent may necessitate the obtainment of more audit evidence to reduce risk to an acceptable level, thereby increasing audit workload costs. Whereas, the misclassification of a fraudulent company as normal could incur high penalties and result in loss of clients.

Reference [8] summarized the misclassification costs for different entities. When a nonfraudulent company is incorrectly predicted as fraudulent (false positive), the regulators bear the costs of the market value loss for the investigated company and the costs of the investigation itself. Further, investors are plagued with the incremental audit fees and opportunity costs of not investing. In addition, auditors incur increased workload and discovery costs. Consequently, plaintiffs' lawyers are more likely to argue subsequent fraud cases more effectively [66]. Conversely, when a fraudulent company is incorrectly classified as nonfraudulent (false negative), regulators face the risk of investors losing confidence in the capital markets. Further, investors suffer financial losses from their investments, whereas auditors face litigation costs, reputational damage, and potential loss of clients. Reference [8] estimated the costs associated with these misclassifications. They found that a high number of FP rendered the predictive models developed in prior studies prohibitively expensive for auditors. This may explain the reason auditors are reluctant to adopt these models in practice.

VI. RESEARCH CHALLENGES AND OPPORTUNITIES

Scholars have invested significant effort in researching FSF prediction models and have reported substantial results. However, several challenges, such as imbalanced FSF samples, cross-period fraud, and hyperparameter tuning, are yet to be solved. These key issues must be addressed.

First, the issue of imbalanced FSF samples must be appropriately addressed. Owing to the low incidence of FSF, its prediction is analogous to finding a needle in a haystack [24]. Sample imbalance severely affects the training and performance evaluation of the model, while hindering its ability to generalize to new data [32]. Four methods to address sample imbalance have been proposed till date. Future research can consider the introduction of other algorithms such as transfer learning. Transfer learning applies knowledge from one domain to enhance learning in another. In FSF prediction, the transfer of knowledge from similar datasets in other regions boosts the model's learning from FSF samples. Consequently, the performance and stability are enhanced.

Second, the addressal of cross-period fraud is crucial in FSF prediction models. This is because FSF often spans multiple periods. Failing to address cross-period fraud can result in feature leakage, thereby impacting model performance and generalization. Although [23] and [24] addressed this issue, the discussion remains inadequate. Many studies have ignored the significance of cross-period fraud or have failed to prioritize it in their research. Future FSF prediction model research should prioritize cross-period fraud by integrating it into model construction and evaluation. Through a comprehensive study of the characteristics and mechanisms of cross-period fraud with the aim of obtaining more effective solutions, the robustness and predictive power of the models can be improved.

Third, the model hyperparameters must be determined to perform model tuning. Prior to model training, the hyperparameters must be determined. These cannot be inferred from the training data and must instead be manually configured. These include factors such as the quantity of hidden layers in a deep neural network model, count of neurons in each hidden layer, and learning rate. Hyperparameter tuning necessitates trials of different parameter combinations to determine the optimal model performance. At times, the choice of hyperparameters impacts the model performance more than the algorithm itself. However, determining the best combination of hyperparameters is not easy. For example, in a deep neural network model, if we desire to attempt different numbers of hidden layers (2, 3, 4), different numbers of neurons per hidden layer (100, 200, 400), different learning rates (0.001, 0.01, 0.1), and different regularization parameters (0.001, 0.01, 0.1), the number of models requiring evaluation in a grid search would reach 81. The consideration of more hyperparameters and possible values results in an exponential increase in the computational load of grid search, thereby requiring more computational resources. This poses a significant challenge. Therefore, determining the appropriate hyperparameters requires substantial computational resources [67]. However, the choice of hyperparameters, such as the number of hidden layers and learning rate, is critical to the performance of the model [68]. In fact, in certain cases, the impact of different hyperparameter combinations on model performance can even surpass

that of the algorithm itself [34]. Researchers can leverage the hyperparameter search functions of machine learning frameworks and employ high-performance computing servers to accelerate the research process.

Fourth, an appropriate code review is necessary. Although machine learning models in the fields of financial accounting and finance have become easier to use and can be developed using a few lines of code, even scholars with a computer science background may make errors when writing code. Logical errors in the code can reduce the convergence speed of the model, thereby increasing the training time and cost. Code errors can also result in the model falling to a local optimum, thereby preventing it from achieving the global optimum. Consequently, the generalization ability and prediction accuracy would be affected. Moreover, logical errors can bias the evaluation of model performance, resulting in incorrect model evaluation results. Reference [23] had an error in their code that improved model performance to a certain extent; the authors promptly corrected it and published an erratum. Therefore, proper code review is a necessary step for ensuring the correctness of the code during the research process.

Fifth, the appropriate use of the performance metrics of FSF prediction models in a scientific and reasonable manner based on the user's purpose is essential. As mentioned earlier, precision and recall usually exhibit an inverse relationship, and solely pursuing the improvement of one metric is not always appropriate. Economic agents must weigh costs and benefits in their decision-making processes [69], [70]. The costs associated with different types of misclassification vary. Consequently, different decision-makers bear different costs owing to these misclassifications [8]. FSF prediction models are both academic research and practical application issues. Therefore, it is necessary to consider the performance metrics of the model for different target users and purposes, thereby balancing the costs and consequences of different types of misclassifications to select the appropriate evaluation metrics and training strategies. Considering the varying needs and preferences of different decision-makers for FSF prediction models, multiple prediction models tailored to different users can be developed. These can provide more detailed prediction explanations.

Sixth, the exploration of FSF prediction models leveraging large language models (LLMs) must be encouraged. Owing to their vast parameters and complexity, LLMs necessitate substantial data for effective training. In addition to traditional financial data, various internet resources such as social media, news, industry reports, and company websites provide crucial data for model training. These internet resources, which encompass diverse FSF event aspects, are vital for enhancing model training and performance. For example, [71] introduced an LLM-based model to detect financial misinformation effectively. Their study demonstrated that LLMs significantly boost financial fraud detection accuracy, particularly in case of identifying social media and news-based

fraudulent behaviors. The integration of internet resources with company financial data for model training is a key research area in FSF prediction.

VII. WHAT CAN INDUSTRY ASSOCIATIONS DO?

In reviewing the research on FSF prediction, this study reached the profound conclusion that relying solely on academic efforts is insufficient. Collaboration among scholars, as well as the integration of research outcomes with practical applications or through joint development, is crucial. This is because existing studies, based on various data sources, feature variables, and algorithms, have yielded diverse FSF prediction models. Each research team has presented unique solutions from their own perspectives, often claiming their methods to be more effective. However, the generalizability and practical effectiveness of these models remains questionable.

The key issue is the consolidation of these research findings and translating them into practical tools that can be utilized by regulators, accounting firms, analysts, and investors. Based on the analysis of research challenges and opportunities, the study further propose specific actions that should be implemented by industry associations. These associations should include not only professional organizations among scholars but also collaborative entities that bridge academia and practice.

A. DEVELOP A SHARED AND OPEN FSF DATABASE

The availability and reproducibility of data are the foundations for ensuring research reliability and dissemination. However, even when literature describes data sources, collection procedures, and processing steps comprehensively, subsequent researchers are still plagued by many challenges when attempting to replicate the results. First, despite detailed descriptions of each data processing step in the literature, the actual implementation may depend on specific contexts, tools, or proprietary knowledge that are difficult to convey fully in writing. For example, the data screening criteria, sequence of data cleaning steps, or feature transformation algorithms may not be thoroughly explained in the literature. Moreover, the descriptions may lack sufficient detail, thereby rendering replication difficult. Second, even with comprehensive descriptions, different subsequent researchers may interpret and implement the same processing steps differently owing to variations in experience or understanding. This would result in subtle differences from the original study that can affect the results. Finally, in certain cases, the original authors may omit certain steps that appear unimportant but are crucial to data processing. These omissions could be unintentional or because certain steps are considered routine; however, subsequent researchers may not completely grasp these practices.

Although certain studies have made the data and code public, this is far from sufficient. This is primarily because scholars often prioritize the publishing of papers over constructing databases. To enhance the quality and credibility of

their research, scholars may make relevant data and source code available such that other researchers can verify and replicate their results. However, the motivation for sharing data often involves supporting their own research findings, rather than contributing to a shared database. Even when scholars are willing to share data, various reasons, such as data copyright or unique data considerations, may cause them to release only a part of the data, which does not constitute a complete database. Moreover, after rendering data public, scholars are not required to continually update it. Over time, FSF behavior may change, resulting in the emergence of new methods and patterns. FSF prediction models must be based on the latest and most comprehensive data, which static datasets cannot provide. In addition, in the era of information explosion, data is not only massive but also originates from diverse and widespread sources, scattered across different institutions. The integration of these dispersed data remains a challenge [72].

A shared and open FSF database can provide scholars with a unified data platform, significantly simplifying the workload of data collection and organization. It can save time and effort, thereby enabling scholars to concentrate on building prediction models. Moreover, it will significantly boost data reproducibility and model comparability in FSF prediction. A public FSF database facilitates the comparison of study results and verification of prior findings. Continuous collaboration and discussion drive progress in FSF prediction research. The database is crucial for FSF prediction research and serves as a public resource. Current systems lack the effective incentives for scholars to share and open data. Industry associations, as key players in FSF prediction research, play a significant role. They can create dedicated organizations, set standards and requirements for database use, and offer incentives to guide the development and updating of a shared and open FSF database. Further, they can coordinate resources for collecting, organizing, and updating FSF data. This will ensure its timeliness, accuracy, and completeness, while also encouraging data companies to participate in database development. Data companies, leveraging their financial resources and technical expertise, can support and assist in database construction and updates.

Industry associations can create survey tools, such as those by [73], to gain a deeper understanding of FSF. These tools can aid in identifying economic losses from fraud and provide data for the analysis of misclassification costs. The collection of diverse fraud-related data with these tools aids in risk assessment and model development, thereby enhancing FSF detection efficiency and accuracy.

B. HOLD FSF PREDICTION COMPETITIONS

In FSF prediction, a tight link between academic research and practical use is vital. The thorough research of scholars lays a strong theoretical groundwork. Consequently, practical application turns these theories into working prediction models. Although closely related, there exists a gap between academic

research and practical application. Academic research delves into FSF prediction theories and methods, often highlighting theoretical framework building and model refinement. Whereas, practical application is centered on the resolution of real business challenges, thereby emphasizing usability. Thus, finding a balance between academic research and practical application is crucial for advancing FSF prediction.

The organization of FSF prediction contests is a key step in blending academic research with practical application. Collaborations on problems and sharing experiences and technologies among participants helps propel the development and use of FSF prediction. The online platform named Kaggle facilitates numerous data science contests and initiatives across domains such as NLP and image recognition. On Kaggle, participants can access datasets, submit predictions, and compete globally to tackle real-world challenges. An example is the IEEE-CIS Fraud Detection competition in 2019, which is targeted toward spotting fraudulent online transactions. Using historical transaction data and features, participants construct models to predict fraudulent transactions. Vesta supplied the competition with a dataset of nearly 590,000 records, offering a \$20,000 prize. The competition drew 27,053 participants, receiving 125,219 submissions. Companies often sponsor these competitions, consequently providing datasets and significant prize money.

The hosting of FSF prediction contests by industry associations fosters academic-industry integration and facilitates the creation of a collaborative platform. These competitions bring together scholars, experts, and students from diverse backgrounds with the purpose of sharing research, technical insights, and best practices. They broaden horizons, spark innovative thinking, and drive progress in FSF prediction technology. Moreover, competitors can utilize the FSF database for various innovative research projects during the competition. They can experiment with prediction models, feature engineering, and data processing to determine the best solutions. In addition, industry associations often publish top solutions as academic articles. The sharing of top results broadens reach and inspires scholars to pursue further FSF prediction research, thereby fostering enthusiasm and innovation.

VIII. CONCLUSION

The area of FSF prediction is intricate and constantly growing. It poses several challenges and prospects for researchers, practitioners, and those in charge of regulations. A major hurdle is the creation of reliable prediction models. These models must tackle problems such as unequal sample sizes and fraud across different periods, while also avoiding model biases owing to coding errors. Moreover, researchers must focus more on data handling, feature creation, and the assessment of model effectiveness to guarantee their precision and dependability. The growing use of unstructured data in predictions complicates research; however, it also provides fresh avenues for enhancing model effectiveness and forecasting abilities.

Despite these obstacles, leveraging various data sources, deep learning, and LLMs can significantly boost prediction model performance, rendering them more practical for real-life situations. However, to progress in this field, academia and industry must collaborate more closely. Key steps include the creation of shared FSF databases and organizing prediction contests, thereby ensuring research has both theoretical significance and practical use.

With the financial markets continuing to grow more intricately, FSF detection and prevention methods must adapt appropriately. By tackling challenges and capitalizing on opportunities, scholars and practitioners can make strides in the creation of better FSF prediction models, thereby aiding capital market stability and growth.

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